"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. ....
18OMEC1186

B.E. II (M)

3 **Kine of M/c**

SECOND B.E. (MECHANICAL ENGINEERING)

III SEMESTER

EXAMINATION - 2019

ME-203 A: KINEMATICS OF MACHINES (M)

Time - Three Hours

Maximum Marks - 100

Note:- (1) Answer any **FIVE** questions.

(2) All questions carry equal marks.

(3) Any missing data may be suitable assumed and stated clearly.

(4) Q.No. 1 is Compulsory & Complete solution (including Calculations) for Q.No. is to be given in the drawing sheet.

1. The quick return mechanism an shown in figure having the crank and slotted lever type shaping machine. The dimensions of the various link are as follows:

CA = 200mm, AB=100mm, OC=400mm and CR=150mm,

The driving crank AB makes 120° with the vertical and rotates at 120 rpm in the clockwise direction.

Find
1. Velocity of ram R
2. Angular velocity of the slotted link OC.
3. The angular acceleration of the slotted lever. (20)

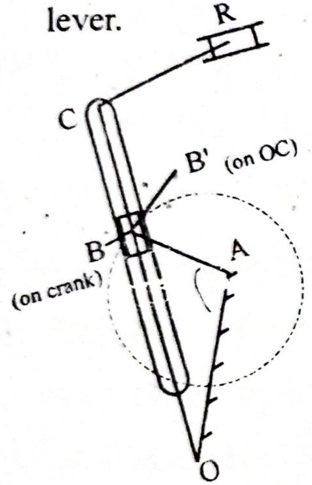

Figure . 1

2. (a) What is the difference between the slider-crank chain and the double slider-crank chain? Give diagrammatic sketches of three mechanisms which are inversions of each of the above chains and state the purpose for which each mechanism is used. (10)

(b) Explain Grubler's criterion for determing degree of freedom in mechanism. (5)

(c) In a crank and slotted lever quick return notion mechanism, the distance between the fixed centres is 240mm and length of the driving crank is 120mm. Find the inclination of the slotted bar with the vertical in the extreme position and time ratio of cutting stroke to the return stroke. (5)

3. (a) Design a pantograph for an indicator to obtain the indicator diagram of an engine. The distance from the tracing point of the indicator is 100mm. The indicator diagram should represent four times the gas pressure inside the cylinder of an engine. (10).

(b) Sketch two different forms of straight-line motion which are based on the four-bar kinematic chain. Show how the best position for the tracing point may be found. (10)

4. (a) Calculate the centrifugal tension in a belt which runs over two pulleys at a speed of 5500 ft/min. The belt is 8 inche wide and 5/16 inche thick, and weighs 0.035 lb/ in^3 If the belt embraces an angle 165°, μ is 0.25 and the maximum permissible stress in the belt material is 350 lb/in^2, Calculate the maximum horse power transmitted at the above speed. (10)

(b) Derive the relation, $\frac{T_1}{T_2} = e^{\mu\theta}$ for a flat - belt drive with usual rotation. (10)

5. (a) Find expression for the cerew efficiency of a square thread. Also, determine the condition for maximum efficiency. (10)

(b) A shaft carring a load of 10 tonnes and running at 120 rpm has a number of collars integral with it. Shaft diameter is 240mm and the external diameter of the collars is 360mm. Intensity of uniform pressure is 400KN/m^2 and the coefficient of friction is 0.06. Determine the power absorbed in overcoming the friction and the number of collars required. (10)

6. (a) Describe the procedure to draw velocity and acceleration diagrams of a four-link mechanism. In what way are the angular acceleration of the output link and the coupler found? (10)

(b) Explain the procedure to construct Klein's construction to determine the velocity and acceleration of a slider-crank mechanism. (5)

(c) The engine mechanism shown in figure has crank OB=50mm and the length at connecting rod AB=225mm. The engine speed is 200rpm.

Find 1. the velocity of slider A and AB.

2. the angular acceleration of AB. (5)

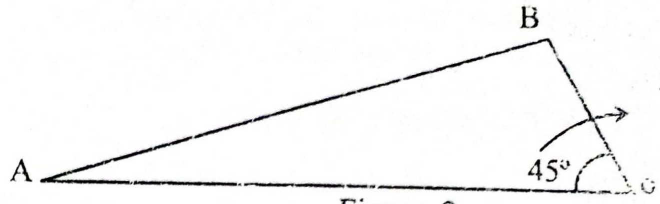

Figure. 2

7. (a) What is meant by cross-belt drive? Find the length of a cross belt drive. (10)

(b) Determine the mechanical efficiency of a wedge used to raise loads if the angle of wedge in 20° and the coefficient of friction is 0.2 between the frame and the wedge and 0.15 between the slider and the guide. The height of the guide is 120mm and its lower end is 45mm above the lower point of the slider which has a width of 50 mm. (10)

6. (a) Derive the expression for maximum power transmitted by the belt drive. 5

(b) 2 KW of Power is transmitted by an Open belt drive. The Linear Velocity of the belt is [illegible] m/s. The angle of lap on the Smaller Pulley is 160°. The Coefficient of Friction is 0.25.

Determine the effect on Power transmission in the following Cases:-

(i) Initial tension in the belt is increased by 8%

(ii) Initial tension in the belt is decreased by 8%

(iii) Angle of Lap is increased by 8% by the use of an idlear Pulley, for the same speed and the tension on the tight side.

(iv) Coefficient of friction is increased by 8% by suitable dressing the friction surface of the belt. 15

7. (a) Discuss relative merits and demerits of belt. rope and chain drive for transmission of Power. 10

(b) A Plate Clutch has three discs on driving Shaft and two discs on the driven Shaft, Providing four pairs of Contact Surfaces. The outside diameter of the Contact Surfaces is 240 mm and inside diameter 120 mm. Assuming uniform pressure and μ =0.3. find the total spring load pressing the plates together to transmit 25 KW at 1575 r.p.m.

if there are 6 Springs each of Stiffness 13 KN/M and each of the contact Surfaces has Worn away by 1.25 mm, find the maximum power that can be transmitted, assuming Uniform Wear. 10

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Roll No. ...16UMEC1118...

B.E. II (M)

3 **Kine of M/c**

Second B.E. Mechanical Engineering
III SEMESTER
EXAMINATION-2018
ME-203 A: KINEMATICS OF MACHINES (M)

Time - Three Hours

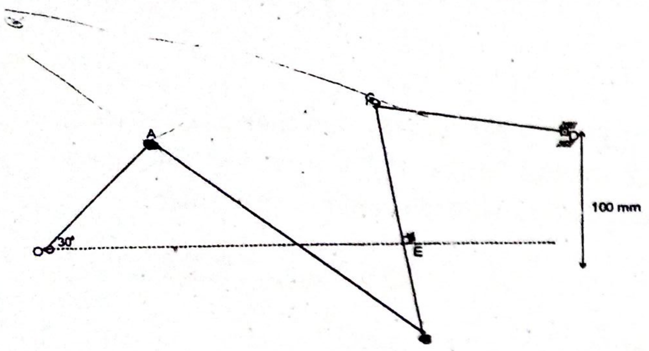
Maximum Marks - 100

Note :- (1) Answer any FIVE questions.

(2) All Questions Carry Equal marks.

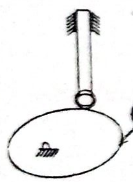
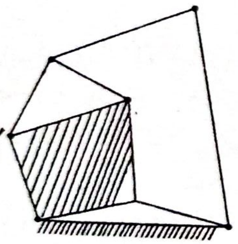
(3) Any missing data may be suitably assumed and stated clearly.

(4) Q.No.1 is Compulsory & Complete solution (including Calculations) for Q.No. is to be given in the drawing sheet.



OA=200 mm, OE=1.35 m, Ab=1.5 m, BE=400 mm BC=600 mm, CD=500 mm and AOE=30° In the given diagram, if Link OA rotates at 120 rpm in clockwise direction. then find out the linear velocity and angular velocities of all the Links Using instantaneous Centre method. 20

2. (a) Determine the degree of freedom of the given systems:

(i)  (ii)  (5)

(b) Explain the different types of relative motions in mechanisms with suitable examples. 5

(c) Describe the Crank and slotted lever quick return motion mechanism. Derive the expression for the quick return ration and the length of Stroke. 10

3. (a) Derive an expression for the ratio of Shafts velocities for Hooke's Joint and draw the Polar

diagram depicting the Silent features of driven shaft Speed. 10

(b) What are the Straight Line mechanisms? Describe one type of exact straight Line Motion mechanism with the help of a Sketch. 10

4. (a) Select one of the following two bearings and Justify why did you reject the other?

(i) Deep groove ball bearing:-
Dynamic Load Capacity=47,000 N
Radian Factor (x)=0.56, Axial Factor (Y)=1.83

(ii) Cylindrical roller bearing:
Dynamic load Capacity= 28,000 N
Radial Factor (x)=1.0, Axial Factor (Y)=0.0
Ration Factor (v)=1, Reliability factor=0.62 Radial load=3000 N and axial load=1600N and RPM=1000 10

(b) Derive an expression for the frictional torque for a flat collar bearing in terms of inner radius r_1 outer radius r_2, axial thrust W and coefficient of friction μ. Assume (i) Uniform pressure (ii) uniform Wear. 10

5. (a) Write a short note on Frendentein's methods of synthesis.? 10

(b) What is meant by the expression " friction circle"? Deduce an expression for the radius of friction Circle in terms of the radius of the Journal and the angle of Friction. 10

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Roll No. 1AMEC38549

B.E. II (M)

3 **Kine. of M/c.**

SECOND B.E. MECHANICAL ENGINEERING

III SEMESTER EXAMINATION - 2017

ME-203 A : KINEMATICS OF MACHINES (M)

Time - Three Hours

Maximum Marks : 100

Note : - (1) Answer any **Five** questions.

(2) All questions carry equal marks.

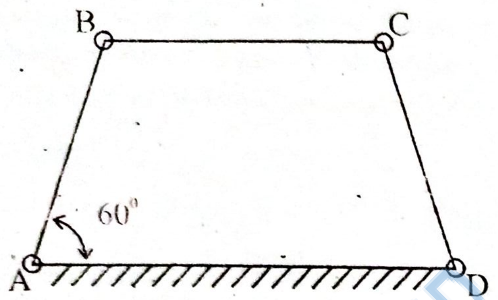
(3) Any missing data may be suitably assumed and stated clearly.

(4) Q.No.1 is compulsary and complete solution(including calculations) for Q. No.1 is to be given on the drawing sheet.

(3)

1. Figure shows the configuration diagram of a four bar mechanism along with the lengths of the links in mm. The link AB has an instantaneous angular velocity of 10.5 rad/s and a retardation of 26 rod/s^2 in the counter-clockwise direction. Find

(i) The angular acceleration of links BC and CD



$\angle BAD = 60^\circ$. AB=50mm, BC=66mm, CD=56mm, AD=100mm 20

2. (a) How are the kinematic pairs classified. Explain with examples. 8

(b) What do you mean by degree of freedom of a kinematic pair? How it is determined? 6

(c) Define Grashof's law. State how it is helpful in classifying the four link mechanism into different types. 6

3. (a) Describe various inversions of a four bar link mechanism giving examples.. 12

(b) What is fundamental equation of steering gears? Which steering gear fulfills this condition. 8

4. What is the coriolis acceleration component? In which case does it occur? How it is determined. 20

5. (a) What is friction? It it a blessing or curse? Justify your answers giving examples. 6

(b) What is uniform pressure and uniform wear theories. 6

(c) What is a clutch? Make a sketch of a single plate clutch and describe its working. 8

6. A V- belt drive with the following data transmits power from an electric motor to a compressor:

Power transmitted	= 100kW
Speed of the electric motor	= 750 rpm
Speed of the compressor	= 300 rpm
Diameter of compressor pulley	= 800 mm
Centre distance between pulleys	= 1500 mm
Max. speed of the belt	= 30 m/s
Mass density of the belt	= 900kg/m^3
Cross-sectional area of the belt	= 350 mm^2
Allowable stress in the belt	= 2.2 N/mm^2
Groove angle of the pulley	= 38°
Coefficient of friction	= 0.28

Determine the

(a) Length of each belt

(b) Number of belt required

(c) Torque on each shaft 20

7. Write short notes on any two :

(a) Ball and roller bearings

(b) Mechanism for straight line

(c) Chain drives

(d) Frendentein's methods of synthesis. 2×10

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Roll No. ...15 MEC ...

B.E. II (M)

3 **Kine. of M/c.**

SECOND B.E. EXAMINATION 2016

(Mechanical Engineering)

(III-SEMESTER EXAMINATION SCHEME)

ME-203 A : KINEMATICS OF MACHINE (M)

Time Allowed - Three Hours

Maximum Marks - 60

Note : (1) Attempt any **FIVE** Questions.

(2) All Questions carry Equal marks.

(3) Any missing data may be suitably assumed and stated clearly.

(4) Q.No. 1 is compulsory and complete solution (including calculations) for Q.No. 1 is to be given on the drawing sheet.

Q1. A link AB of a four bar linkage ABCD revolves uniformly at 120 rpm in a clockwise direction. Find the angular acceleration of links BC and CD.

Given AB = 75mm   BC = 175 mm

CD = 150 mm   DA = 100mm (fixed)

and $\angle BAD = 90^0$ 12

Q2. a. What do you mean by inversion of a mechanism? Explain with sketches all the inversions of a single slider crank mechanism. Where these inversions are used?   6

b. What do you mean by constrained motions? What are the different types of constrained motion? Explain each type with example and neat sketches.   6

Q3. a. What do you mean by straight line mechanism? Name the mechanisms which are used for straight line motion with examples.   6

b. What is an automobile steering gear? Which steering gear is preferred and why?   6

Q4. a. What are the various kinds of friction? Discuss each in brief.   4

b. In a trust bearing, the external and the internal diameters of the contacting surfaces are 320 mm and 200mm respectively. The total axial load is 80KN and the

intensity of pressure is 350 KN/m^2, The shaft rotates at 400 rpm. Taking the coefficient of friction as 0.06, calculate the power lost in overcoming the friction. Also, find the number of collars required for the bearing.   8

Q5. An open belt drive connects two pulleys 1200mm and 500mm diameters, on parallel shafts 4000mm apart. The maximum tension in the belt is 1855 N. The coefficient of friction is 0.3. The driver pulley of diameter 1200mm runs at 200rpm. Calculate

(i) the power transmitted, and

(ii) torque on each of the two shafts.   12

Q6. (a) Derive an expression for the length of an open belt.   8

(b) What are the relative advantages & disadvantages of chain and belt drives?   4

Q7. **Write short notes on any two:**   **2×6**

(a) Single and multiple disc clutches.

(b) Chain drives

(c) Kinematic synthesis of mechanism.

(d) Coriolis component of acceleration.

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Roll No. 12MEC42076

B.E. II (M)

3 **Kine. of M/c.**

Second B.E. Examination, 2015

(Mechanical Engineering)

(III-Semester Examination Scheme)

ME–203 A : KINEMATICS OF MACHINE (M)

Time—Three Hours

Maximum Marks—60

Note :—(1) Attempt any **FIVE** questions.

(2) All questions carry equal marks.

(3) Any missing data may be suitably assumed and stated clearly.

(4) Q.No. 1 is compulsory and complete solution (including calculations) for Q.No. 1 is to be given on the drawing sheet.

1. The crank of a slider crank mechanism is 15 cm and the connecting rod is 60 cm long. The crank makes 300 rpm in the clockwise direction. When it has turned

45° from the IDC position, determine :

(i) Acceleration of the mid point of the connecting rod.

(ii) Angular acceleration of the connecting rod. 12

2. Explain why Hooke's joint is used to transmit motion from the engine to the differential of an automobile. Also for Hooke's joint connecting the driving and driven shaft prove that speeds of these two shafts will be equal if $\tan\theta = \pm\sqrt{\cos\alpha}$

where, α – angle of inclination of driven shaft with driving shaft

θ – angle turned by driving shaft at any instant. 12

3. (a) Describe various inversions of a four bar link mechanism giving examples. 6

(b) What do you understand by degree of freedom ? Explain and derive Grubler's equation. 6

4. (a) Describe with a neat sketch the working of a cone clutch. 4

(b) What do you mean by greasy friction ? Describe in detail the greasy friction of a journal and deduce the expression for friction circle. 8

5. (a) What do you mean by 'Chordal action' ? Explain its effect on the performance of a chain drive. 5

(b) Derive the expression for optimum speed of flat belt for the transmission of maximum power considering the effect of centrifugal tension. 5

(c) Differentiate between initial tension and centrifugal tension in a belt. 2

6. Power is transmitted using a V-belt drive. The included angle of V-Groove is 30°. The belt is 2 cm deep and maximum width is 2 cm. If the mass of the belt is 3.5 gm per cm length and maximum allowable stress is 140 N/cm^2. Determine the maximum power transmitted when angle of lap is 140° and $\mu = 0.15$. 12

7. Write short notes on (any **THREE**) :

(a) Coriolis component

(b) Grashof's criterion

(c) Instantaneous centre method

(d) Construction of ropes

(e) Ball and roller bearings. 3×4=12

Allowable stress in the belt	$= 2.2\ N/mm^2$	
Groove angle of the pulley	$= 38°$	
Coefficient of friction	$= 0.28$	

Determine the number of belts required and the length of each belt. (12)

7. Write short notes on **ANY THREE:**

(i) Coriolis component of acceleration

(ii) Frendestein's methods of synthesis

(iii) Ball & roller bearings

(iv) Instantaneous centre method for velocity determination

(v) Design of a 4 bar chain for constant angular velocity ratio of cranks. (3x4=12)

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Roll No. ..3200056.

B.E. II (M)

3

Kine. of M/c

Second B.E. Examination, 2014
(Mechanical Engineering)
(II - Semester Examination Scheme)

ME - 203 A : KINEMATICS OF MACHINE (M)

Time : Three Hours
Maximum Marks : 60

Note : (1) Attempt any **FIVE** questions
(2) All questions carry equal marks.
(3) Any missing data may be suitably assumed and stated clearly.
(4) Q.No. 1 is compulsory and complete solution (including calculations) for Q.No. 1 is to be given on the drawing sheet.

1. A Link AB of a four bar linkage ABCD revolves uniformly at 120 rpm in a clockwise direction, find the angular acceleration of link CD, AD is fixed

Given AB = 75 mm	BC = 175 mm
EC = 50mm	CD = 150 mm
DA = 100 mm	∠BAD = 90°

(12)

(Contd.)

2.(a) Derive the condition constraining a point in a mechanism along an exact straight line path. Describe one form of mechanism composed of turning pairs only that will contain a point in it to move along an exact straight line. (8)

(b) Obtain the condition that must be satisfied by the steering mechanism of an automobile in order that the wheels may have pure rolling motion while rounding a curve. (4)

3.(a) Describe various inversions of a four bar link mechanism giving examples. (4)

(b) How are the kinematic pairs classified. Explain with examples. (3)

(c) What do you understand by degree of freedom? For a plane mechanism, derive an expression for Grubler's equation. (5)

4.(a) Describe with neat sketch the working of a single friction clutch. (4)

(b) What are the laws of solid dry friction?

(c) What is uniform pressure theory? Deduc expression for the friction torque considering any one theories for a flat collar.

5.(a) What do you mean by "Chordal action."? Explain its effect on the performance of a chain drive. (5

(b) Deduce an expression for the exact and approximate lengths of an open belt drive. (7

6. A V-belt drive with the following data transmits power from an electric motor to a compressor:

Power transmitted	= 100 kw
Speed of the electric motor	= 750 rpm
Speed of the compressor	= 300 rpm.
Diameter of compressor pulley	= 800mm
Centre distance between pulleys	= 1500mm
Maximum speed of belt	= 30m/s
Mass density of the belt	= 900 kg/m³
Cross section area of belt	= 350 mm²